The AI code script that was downloaded from GitHub it's whole purpose was to clean up the healthcare CSV and compute summary statistics as well as anonymize and encrypt any data that was in that CSV. In my review I have found three bugs that produced wrong data and several logic and design weaknesses.

The first bug was that the blood types were being corrupted. It was called a strip title as such it changed the names of the blood types. As well the duplication cleanup was running before the normalization even though the code said otherwise. Rose that only differed in capitalization were treated as distinct and survived the purge. Third, the cleaning report under-counted the problems. It only recorded out of range values while entries failed silently becoming missing and never appeared in the report.

Some of the logic and design issues are that the key file defaults to the same folder as the encrypted data. As such anyone who obtains the folder can decrypt it. Patient IDs are flawed as they are plotted on the road position along with the name so one person admitted twice would have two different IDs and every ID changes if the row order changes. Replacing names is only one step of the anonymization step. If the Doctors hospitals and exact dates are not de-identified it can allow for a patient to be identified.

My improvements are that I rewrote the clean data function so now blood type is now uppercase and the new canonical case helper identifies case variance of insurance, medication, doctors and hospitals but keeping them the most common spelling. All normalization now happens before the drop duplicates so case only duplicates are removed. The reports gained unparable talents that

separate unreadable values from individual ones. The fix can be checked by printing blood type value counts which should only show a b AV and o with a plus or minus sign. There are still design issues I would recommend storing the data apart from the data making patient IDs stable across administrations.